

2.1 (a)

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

Sample mean: $\bar{x} = 1$, $\bar{y} = 2$.

$$(b) \quad b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \cdot 1 = 1.2$$

$$(c) \quad \sum_{i=1}^5 x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum_{i=1}^5 x_i y_i = 12 + 4 + 3 - 1 + 0 = 18$$

$$\sum (x_i - \bar{x})^2 = 10$$

$$\sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1^2 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$$

$$\sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8$$

$$\left. \begin{array}{l} \sum (x_i - \bar{x})^2 = 10 \\ \sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1^2 = 10 \end{array} \right\} \Rightarrow \sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2$$

$$\left. \begin{array}{l} \sum (x_i - \bar{x})(y_i - \bar{y}) = 8 \\ \sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8 \end{array} \right\} \Rightarrow \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N \bar{x} \bar{y}$$

(d)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\hat{y}_i = b_1 + b_2 x_i = 1.2 + 0.8 x_i, \text{ and } \hat{e}_i = y_i - \hat{y}_i$$

$$\text{Sample variance of } y: S_y^2 = \sum_{i=1}^N \frac{(y_i - \bar{y})^2}{N-1} = \frac{1}{4}(4+1+0+1+4) = 2.5$$

$$\text{Sample variance of } x: S_x^2 = \sum_{i=1}^N \frac{(x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

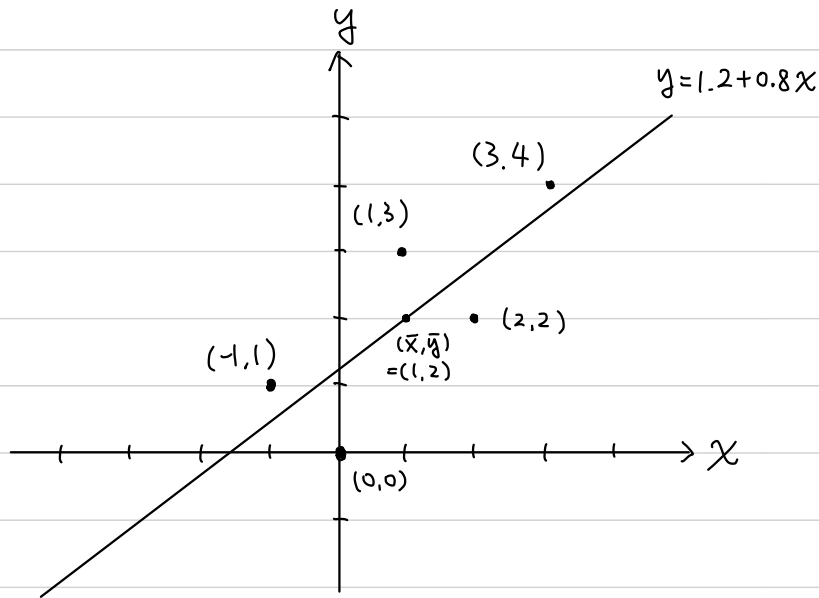
Sample variance between X and y : $S_{xy} = \sum_{i=1}^N \frac{(y_i - \bar{y})(x_i - \bar{x})}{N-1} = \frac{8}{4} = 2$

Sample correlation between X and y : $r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{2.5} = 0.8$

Coefficient of variation of X : $CV_x = 100 \cdot \left(\frac{S_x}{\bar{x}} \right) = 100 \cdot \sqrt{2.5} = 25\sqrt{5}$

Median of x : 1.

(e)



x	y
3	4
2	2
1	3
-1	1
0	0

(g) $\bar{y} = 2$, $b_1 + b_2 \bar{x} = 1.2 + 0.8 \cdot 1 = 2 \Rightarrow \bar{y} = b_1 + b_2 \bar{x}$

(h) $\bar{\hat{y}} = \sum \frac{\hat{y}_i}{N} = \frac{1}{5} \cdot 10 = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $\hat{\text{var}}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$\text{se}(b_2) = \sqrt{\hat{\text{var}}(b_2|x)} = \sqrt{0.12} = \frac{\sqrt{3}}{5}$

2.22

a.

```
> library(POESRdata)
> mod1 <- lm(totalscore ~ small, data = star5_small)
> smod1 <- summary(mod1)
> smod1
```

Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:

Min	1Q	Median	3Q	Max
-191.63	-50.63	-7.12	42.37	324.38

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	918.628	2.536	362.260	<2e-16 ***
small	9.989	4.611	2.166	0.0305 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared: 0.003903, Adjusted R-squared: 0.003071
F-statistic: 4.694 on 1 and 1198 DF, p-value: 0.03047

The estimated coefficient on small (9.989) indicates that student in small classes score about 10 points higher on average than students in regular classes, and this difference is statistically significant at the 5% level ($p=0.0305$).

$$TOTALSCORE_i = 918.628 + 9.989 SMALL_i + e_i$$

b.

```
> mod2 <- lm(readscore ~ small, data = star5_small)
> smod2 <- summary(mod2)
> smod2
```

Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:

Min	1Q	Median	3Q	Max
-76.123	-21.123	-3.356	15.877	192.877

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	434.123	1.044	416.021	< 2e-16 ***
small	5.466	1.897	2.881	0.00403 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared: 0.006882, Adjusted R-squared: 0.006053
F-statistic: 8.301 on 1 and 1198 DF, p-value: 0.004032

The estimated coefficient on small (5.466) indicates that student in small classes score about 5.5 points higher in reading on average than students in regular classes, and this difference is statistically significant at the 1% level ($p=0.004$).

$$READSCORE_i = 434.123 + 5.466 SMALL_i + e_i$$

```
> mod3 <- lm(mathscore ~ small, data = star5_small)
> smod3 <- summary(mod3)
> smod3
```

Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:

Min	1Q	Median	3Q	Max
-145.505	-35.028	-6.505	28.495	141.495

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	484.505	1.677	288.884	<2e-16 ***
small	4.522	3.049	1.483	0.138

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared: 0.001832, Adjusted R-squared: 0.0009992
F-statistic: 2.199 on 1 and 1198 DF, p-value: 0.1383

The estimated coefficient on small (4.522) indicates that student in small classes score about 4.5 points higher in math on average than students in regular classes, but this difference is NOT statistically significant.

$$MATHSCORE_i = 484.502 + 4.522 SMALL_i + e_i$$

The estimates suggest that students in small classes perform better in both reading and math. However, the effect is statistically significant for reading scores but not for math scores.

C.

```
> mod4 <- lm(totalscore ~ aide, data = star5_small)
> smod4 <- summary(mod4)
> smod4
```

Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:

Min	1Q	Median	3Q	Max
-195.14	-52.14	-6.95	40.95	330.86

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	922.145	2.640	349.250	<2e-16 ***
aide	-1.396	4.437	-0.315	0.753

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared: 8.267e-05, Adjusted R-squared: -0.000752
F-statistic: 0.09904 on 1 and 1198 DF, p-value: 0.753

The estimated coefficient on aide (-1.396) indicates that student in a regular classes with a teacher aide score about 1.4 points lower on average than a class without a teacher aide, but this difference is NOT statistically significant.

$$\text{TOTALSCORE} = 922.145 - 1.396 \text{AIDE} + e$$

d.

```
> mod5 <- lm(readscore ~ aide, data = star5_small)
> smod5 <- summary(mod5)
> smod5
```

Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:

Min	1Q	Median	3Q	Max
-77.908	-21.536	-2.908	15.464	191.092

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	435.9084	1.0882	400.585	<2e-16 ***
aide	-0.3719	1.8285	-0.203	0.839

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared: 3.453e-05, Adjusted R-squared: -0.0008002
F-statistic: 0.04137 on 1 and 1198 DF, p-value: 0.8389

The estimated coefficient on aide (-0.3719) indicates that student in a regular classes with a teacher aide score about 0.4 point lower on average in reading than a regular class without a teacher aide, but this difference is NOT statistically significant.

$$\text{READSCORE} = 435.9084 - 0.3719 \text{AIDE} + e'$$

```
> mod6 <- lm(mathscore ~ aide, data = star5_small)
> smod6 <- summary(mod6)
> smod6
```

Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:

Min	1Q	Median	3Q	Max
-147.236	-32.236	-7.212	27.788	140.788

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	486.236	1.744	278.731	<2e-16 ***
aide	-1.024	2.931	-0.349	0.727

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared: 0.0001019, Adjusted R-squared: -0.0007327
F-statistic: 0.1221 on 1 and 1198 DF, p-value: 0.7268

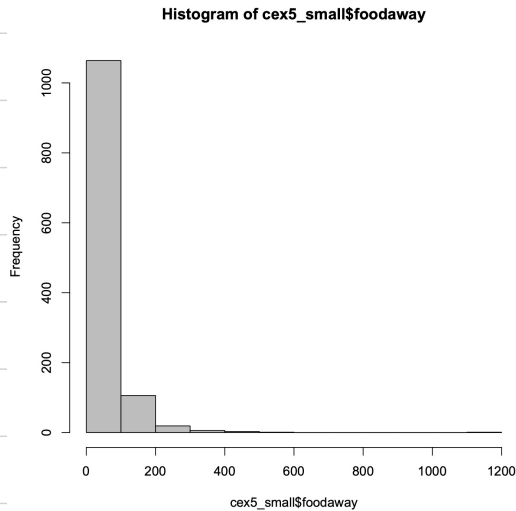
The estimated coefficient on aide (-1.024) indicates that student in a regular class with a teacher aide score about 1 point lower on average in math than a regular class without a teacher aide, but this difference is NOT statistically significant.

$$\text{MATHSCORE} = 486.236 - 1.024 \text{AIDE} + e''$$

The estimates suggest that students in a regular class without a teacher aide perform better than a regular class with a teacher aide in both reading and math, but both of them are not statistically significant.

2.25

a.



← Histogram of FOODAWAY

↓ Summary statistics

```
> library(POE5Rdata)
> data("cex5_small")
> hist(cex5_small$foodaway, col='grey')
> summary(cex5_small$foodaway)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

mean: 49.27 median: 32.55

25th percentiles: 12.04

75th percentiles: 67.50

b. `summary(cex5_small$foodaway[cex5_small$advanced==1])`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	21.67	48.15	73.15	90.00	1179.00

`summary(cex5_small$foodaway[cex5_small$college==1])`

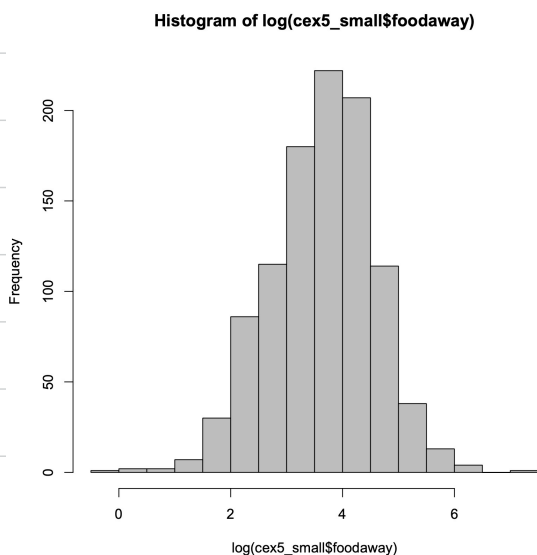
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	14.44	36.11	48.60	68.67	416.11

`summary(cex5_small$foodaway[cex5_small$college==0 & cex5_small$advanced==0])`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.63	26.02	39.01	52.65	437.78

	mean	median
with an advanced degree	73.15	48.15
with a college degree member	48.60	36.11
with no advanced or college degree member	39.01	26.02

c.

`> summary(log(cex5_small$foodaway))`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-Inf	2.488	3.483	-Inf	4.212	7.072

Since at least one data in FOODAWAY is 0, and the log of 0 doesn't exist, so $\ln(\text{FOODAWAY})$ has less observation than FOODAWAY.

d.

```
> mod1 <- lm(log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 0, ])
> smod1 <- summary(mod1)
> smod1
```

Call:

```
lm(formula = log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 0, ])
```

Residuals:

Min	1Q	Median	3Q	Max
-3.6547	-0.5777	0.0530	0.5937	2.7000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1293004	0.0565503	55.34	<2e-16 ***
income	0.0069017	0.0006546	10.54	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom

Multiple R-squared: 0.09826, Adjusted R-squared: 0.09738

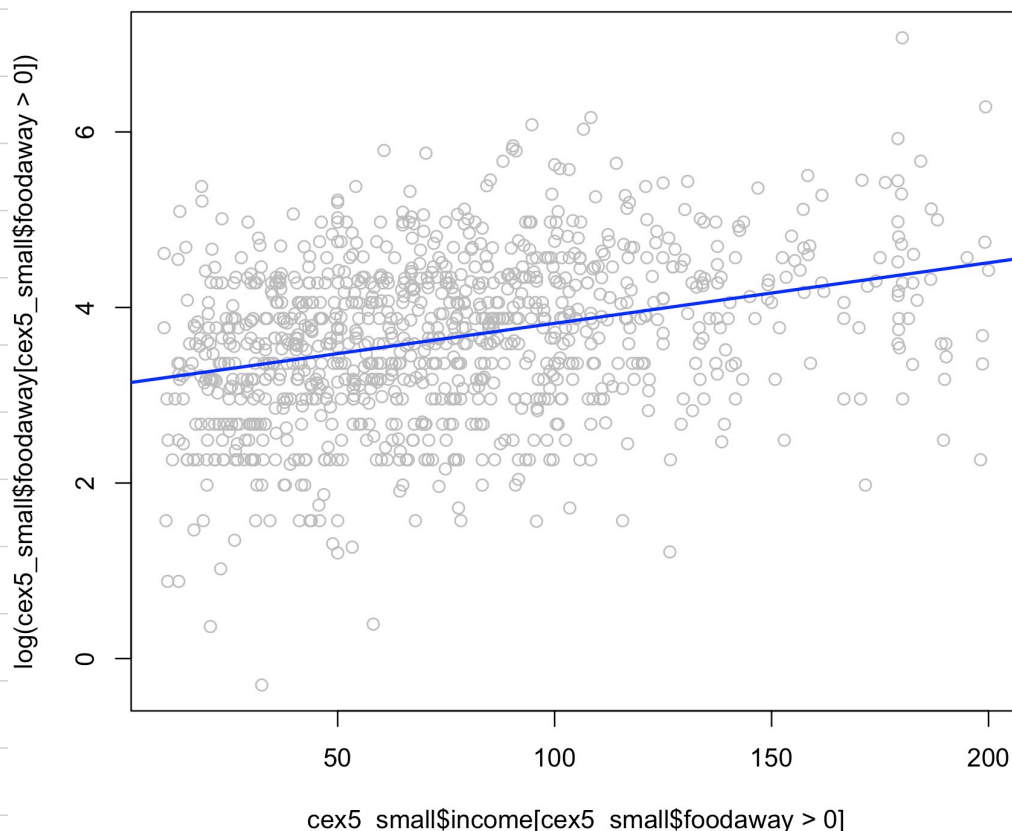
F-statistic: 111.1 on 1 and 1020 DF, p-value: < 2.2e-16

Estimate of the linear regression:

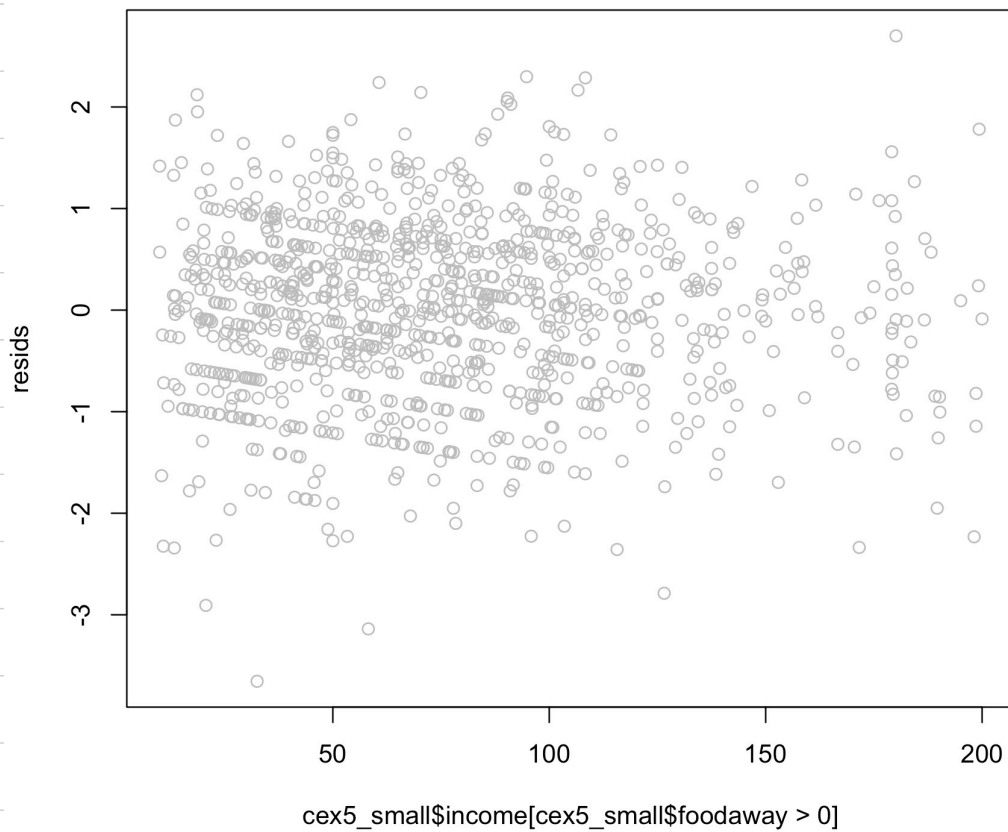
$$\ln(\text{FOODAWAY}) = 3.1293004 + 0.0069017 \cdot \text{INCOME} + e$$

The estimate slope 0.0069017 means that if the monthly income goes up by \$100, the log of the expected food away from home expenditure per month per person will increase approximately by 0.0069017.

e.



f.



The residuals appear to be randomly scattered around zero with no obvious pattern. There's a slight increase in dispersion at higher income levels, suggesting possible mild heteroskedasticity.